

LONG-TERM MEMORY ENGINE FOR AI AGENTS IN RUST

Second Project Review

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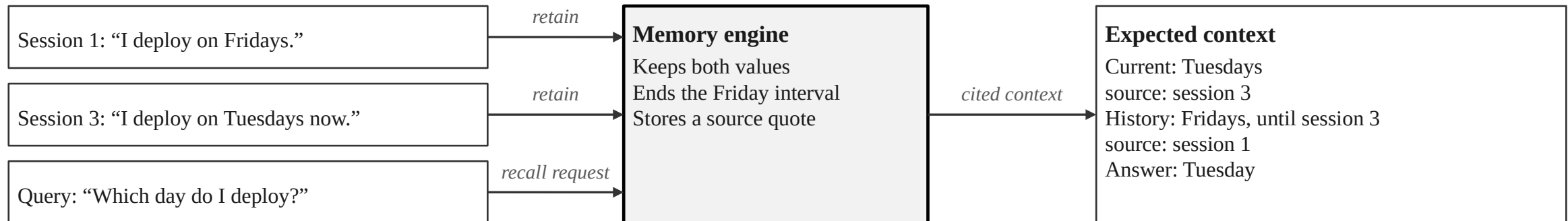
HARI SRINIVASAN, 2023115022

PROBLEM STATEMENT

1 WHY IT IS HARD

- Context limits. A full history costs a lot of tokens, and models tend to miss facts in the middle of a long input [1].
- Forgetting. A new session starts empty unless the old one was stored.
- Corrections. A new value should replace the old one without erasing it.
- Time. A question like “what did I decide last month” needs dated facts [2].
- Provenance. An answer should point to the evidence it came from.

3 WORKED EXAMPLE



2 ENGINEERING STATEMENT

Input: an append-only stream of interaction events (user turns, assistant turns, tool results), each with a timestamp.

Keep: the raw evidence, and every fact derived from it with a validity interval.

Return: at query time, a context within a token budget. Each item names its source evidence and says whether it is current.

LITERATURE SURVEY: MEMORY SYSTEMS

Work and source type	Mechanism	Evidence as reported by the source	Limitation	Relevance here
MemGPT, later Letta, Packer et al., 2023 [3]. Preprint.	Prompt split into main context and external recall and archival storage. The model moves data between them with function calls.	Authors' own tests on OpenAI models. Deep memory retrieval: 92.5% with GPT-4 plus MemGPT against 32.1% for GPT-4 reading a summary.	Memory edits depend on the model's function calling. No latency or token cost figures in the paper.	Origin of memory kept outside the prompt. Here, service code applies the version rules to claims the model extracts.
Zep and Graphiti, Rasmussen et al., 2025 [4]. Preprint, vendor-authored.	Temporal knowledge graph of episodes, entities and facts. Each fact has valid and invalid times, and a contradicting fact ends the old one.	Authors' own LongMemEval run: 71.2% with gpt-4o against 60.2% for the full 115k-token history.	Single configuration, no repeated runs. The authors call their other benchmark (DMR) weak. Needs a graph database.	Closest design for validity intervals. We also run Neo4j, but only as a projection we can rebuild.
Mem0, Chhikara et al., 2025 [5]. Peer reviewed (ECAI 2025), authors are Mem0 staff.	An LLM extracts candidate facts, compares each with the nearest stored memories, then chooses add, update, delete or no-op.	LoCoMo only, authors' run. Judge score 66.9 for Mem0, 72.9 for the full history, which had a p95 latency of 17 s against 1.4 s.	One benchmark and an unnamed judge model. The open-source release has since dropped update, delete and graph memory.	Same extract then compare idea. Here the update rule is code (cardinality and correction flag), not an LLM choice per fact.
Supermemory, Daga et al., 2026 [6]. Company report, not peer reviewed.	Atomic memories linked to source chunks, with updates, extends and derives links and two timestamps.	Reports LongMemEval-S results by category from its own run. Numbers not repeated here.	Engine source closed. The headline depends on the answer model, and the metric "aggregation" is not defined.	Named as a design influence in our repository. Used for mechanism only.
Hindsight, Latimer et al., 2025 [7]. Preprint, vendor-affiliated.	Retain, recall and reflect operations. Recall fuses four retrieval strategies by rank fusion and packs to a token budget.	Reports LongMemEval and LoCoMo results run by the authors. Numbers not repeated here.	Not peer reviewed. No ablation or variance found, and the token budget used is not stated.	Named design influence. We have no reranker.

CHALLENGES AND ISSUES IDENTIFIED

1 WHAT WE DID WRONG IN THE DESIGN

- We treated memory as plain chunks of text and searched them, with no context around each fact.
- Chunk search cannot tell that a new fact replaces an old one, so old and new values came back together.
- Nothing linked related facts, so a question that needs two connected facts could find only one of them.
- We had no way to show where an answer came from.
- We put a small distilled model on the serving path, and it was trained on a format we no longer use.

3 WHAT THE KNOWLEDGE GRAPH FIXED

Facts are nodes with links (supersedes, contradicts, extends). A correction keeps the old fact as history. Every fact points to its exact quote and raw event, and recall follows links to related facts.

Prof. Vijaylaxmi Ma'am's guidance pointed us to the knowledge graph.

2 WHAT WENT WRONG WHEN WE RAN IT

- Memory was per repository, so a session in another directory saw nothing.
- We re-sent the whole chat every turn: 4 turns stored 32 events and only 14 were different.
- A fresh session right after the first got no memory, because we embedded only after extraction.
- A bad model reply was cached, so every retry repeated it and later jobs got stuck behind it.
- The model re-extracted the old Friday value as new, and the current Tuesday value was lost.
- Recall had no distance limit, so about 2,000 tokens of unrelated text went in before almost every prompt.
- The date planner called a model on words like “when”, which took 4.46 s and 19.28 s on two queries.
- The default worker had no credentials, and Ollama silently cut 15 of 18 batches at its 4,096 token default.

PROJECT OBJECTIVES

Objective	Target	Measured	Status
Provenance	100% of stored memories pass an automated provenance check.	323 of 323 facts in user:dev have a source, and every quote is found word for word in its stored text.	Completed
Correct version history	Zero incorrect active versions on held-out update cases.	0 of 106 single-value slots hold two active facts, and the scripted Pi check (told, corrected, asked again) passes. There is no held-out update set yet, and only 2 facts are superseded.	In progress
Small local extractor	Student F1 within 5 points of the teacher at one third of its latency.	Relaxed F1 0.398 for the student and 0.000 for the base model on 136 held-out windows, agreement with the teacher labels. Student p50 latency 6,115 ms against 2,868 ms, so the latency target is not met.	In progress
Retrieval gain	Match a pgvector-only baseline overall and beat it by 5 points on knowledge-update and temporal questions.	Not run. LongMemEval-S needs a 277 MB dataset and a lot of model use.	Planned
Retrieval latency	p95 retrieval at most 300 ms at 10 QPS.	Date planning off: p50 58 ms, p95 72 ms, p99 75 ms (300 requests, 0 errors). With the date planner in the path p95 is 1,870 ms, because 4 of 40 test questions start a model call of 1.6 to 2.1 s.	In progress
Context size	Median returned context at most 1,000 tokens.	Median 1,942 tokens, p95 2,041, with the 2,048 token budget Pi sends (characters divided by four). Target not met.	In progress
Reproducibility	Dataset revision, models, prompts and configuration pinned, raw run outputs published.	Dataset pinned by SHA-256 and settings kept in benchmark/config.json. No run exists yet, so there are no outputs to publish, and the Docker image tag still floats.	In progress
Cross-session memory in Pi (new)	A fact told in one Pi session is recalled in a fresh session in another directory.	14 of 14 scripted checks in each of 4 runs (claude-haiku-4-5 or claude-sonnet-5-5 through Pi). One scenario, one machine.	Completed

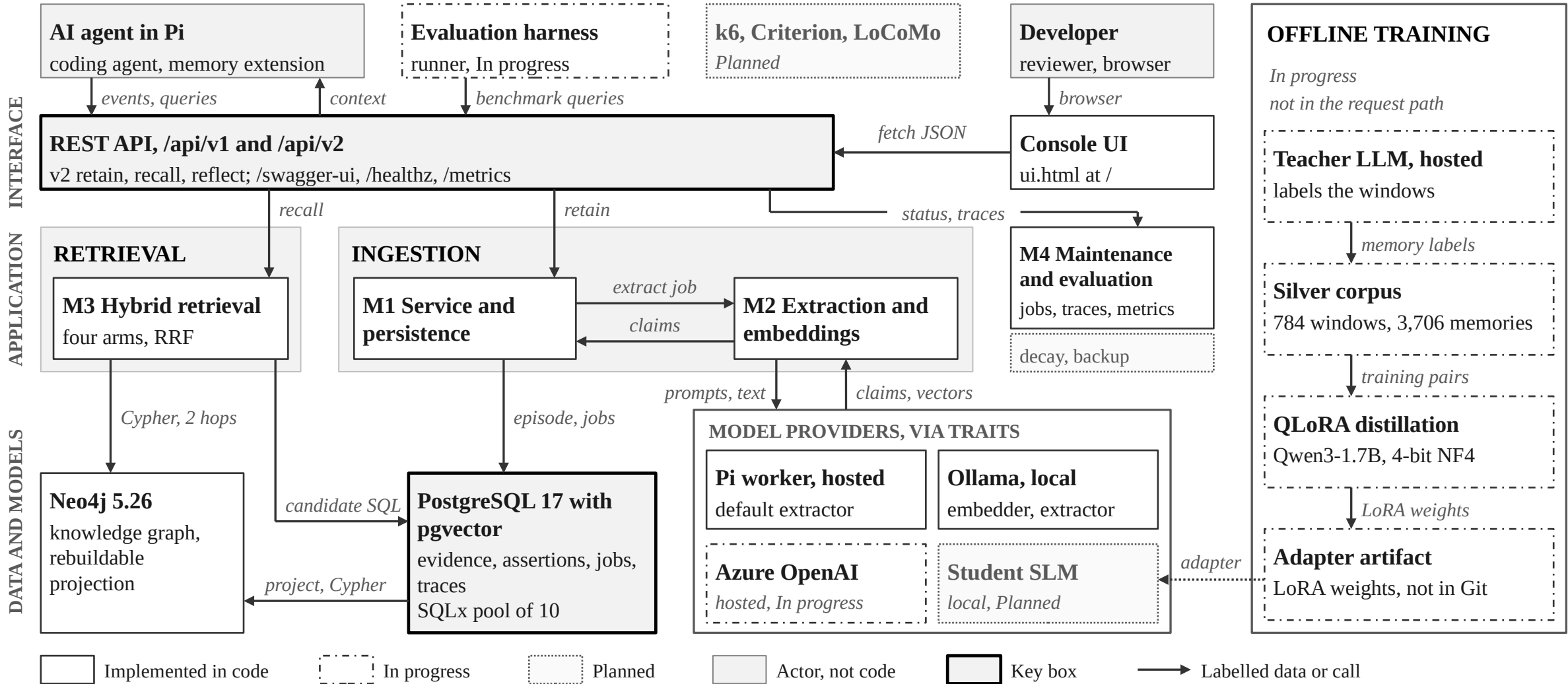
CHANGES SINCE FIRST REVIEW

Area	First Review	Now
Stores	One PostgreSQL 17 database, no graph store	PostgreSQL 17 with pgvector, plus a Neo4j 5.26 graph projection
API	Four routes: session events, request status, context, delete	/api/v1 with 8 routes and /api/v2 with 13 (retain, recall, reflect, jobs, graph)
Write path	One transaction, then 202 and a request id	Retain stores episode, chunks and an extract job in one transaction. Workers extract later.
Extraction	Distilled Qwen3-1.7B student on the serving path	Pi-hosted worker. Optional local Qwen2.5-14B in Ollama, valid on 2 of 10 batches. The student is an offline pipeline, not wired in.
Retrieval	Full-text and vector search, rank fusion	Lexical, semantic, temporal and graph searches, rank fusion (k = 60), raw evidence fallback, optional distance cutoff
Agent integration	None	Pi extension: spool, retain, recall, injected context
Graph view	None	Live page at /graph that draws the memory of one namespace from the real data
Evaluation	LongMemEval-S and LoCoMo, baselines B0 to B3, k6	LongMemEval-S runner, no completed run yet. LoCoMo and k6 are Planned.
Tests	None reported	Rust 83, Python 5 and TypeScript 9, all passing (slide 14)
Observability	Request traces, OpenTelemetry	Traces in PostgreSQL, /healthz, /metrics, JSON logs

Claims that no longer hold

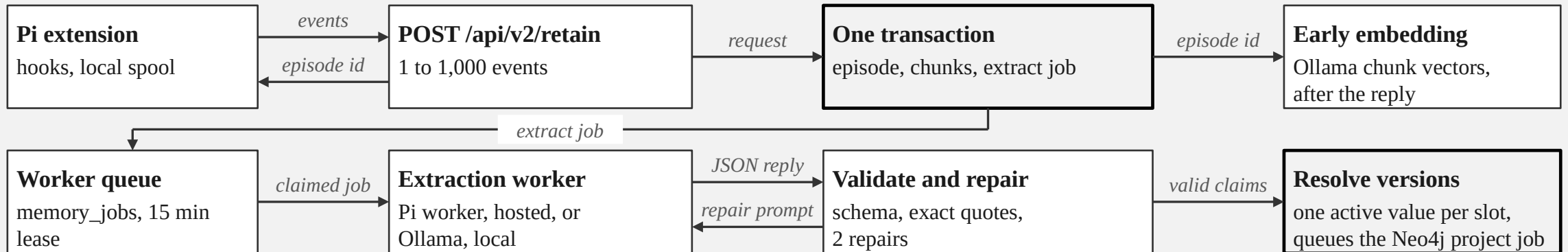
- First Review: no second store. Now: PostgreSQL plus a Neo4j 5.26 graph projection.
- First Review: a distilled student does the extraction. Now: a hosted model through Pi, and the student is not wired in.
- First Review: no memory data leaves the machine. Now: on the default path, event text goes to a hosted model.

REFINED SYSTEM ARCHITECTURE

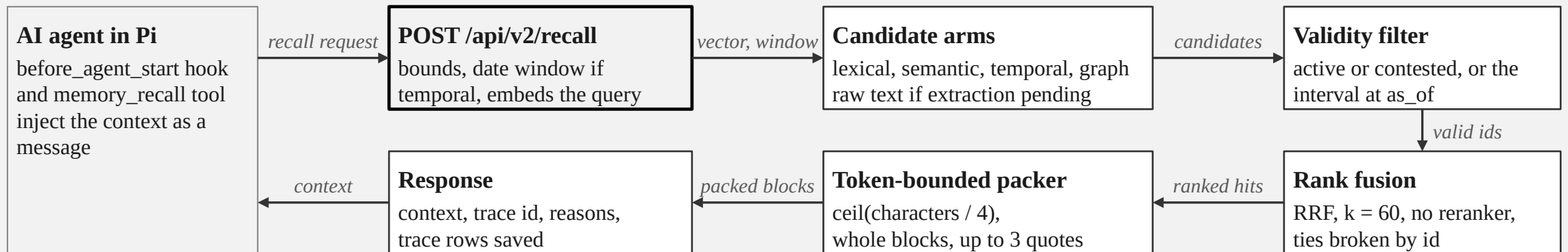


INGESTION AND RETRIEVAL PATHS

1 INGESTION PATH



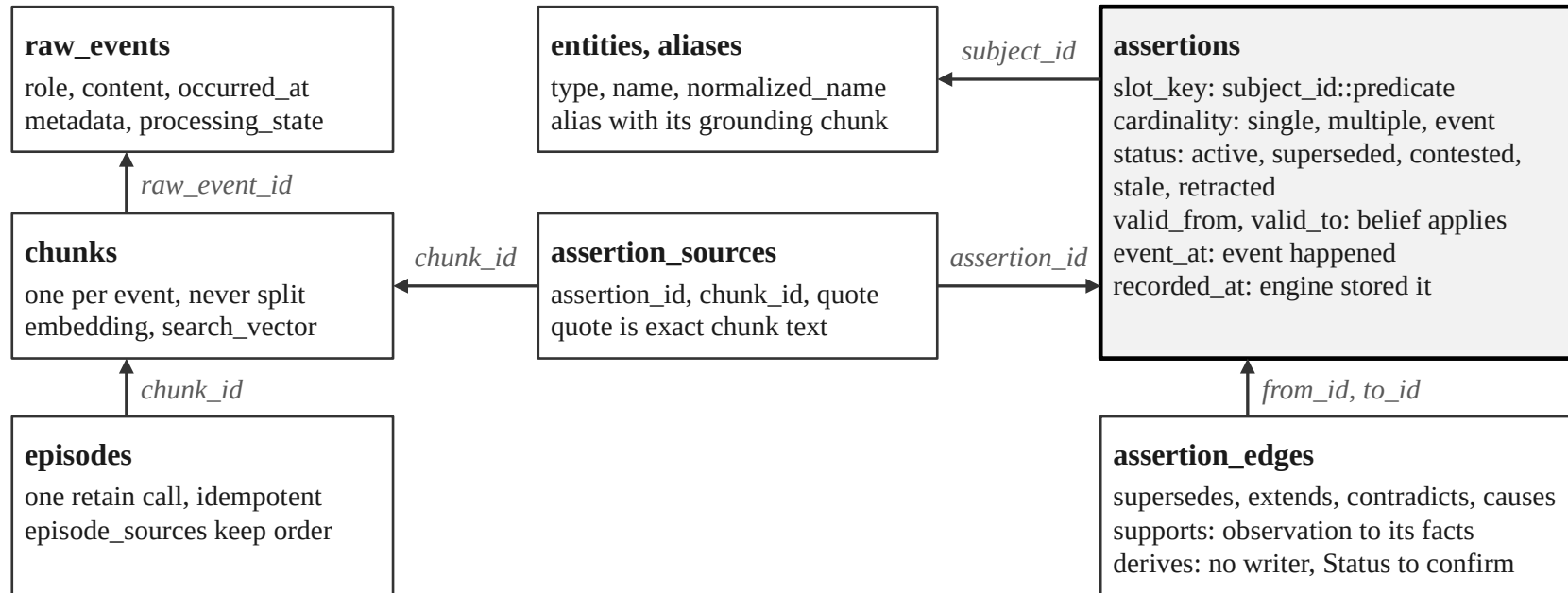
2 RETRIEVAL PATH



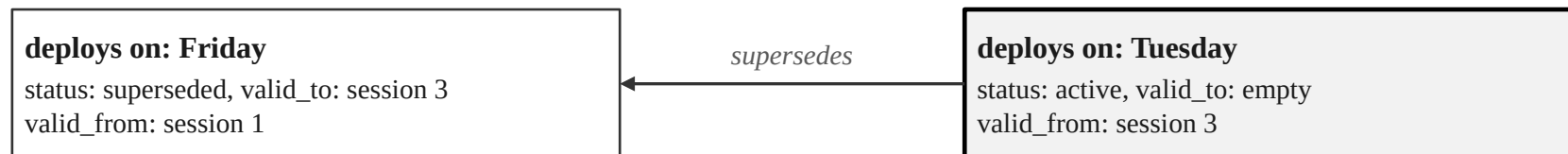
Implemented in code
 Actor, not code
 Key box
 → Labelled data or call

DATA MODEL AND KNOWLEDGE GRAPH

1 TABLES AND LINKS



2 VERSION CHAIN FOR ONE SLOT



3 GRAPH PROJECTION

PostgreSQL 17, authoritative

Entities resolve by exact type and name, else one grounded alias, never by similarity. Every update raises a revision.

↓ *assertion id, revision*

Neo4j 5.26, projection

Assertion, KnowledgeEntity and MemoryNamespace nodes, MENTIONS and SUPPORTS edges. A write with an older revision is dropped.

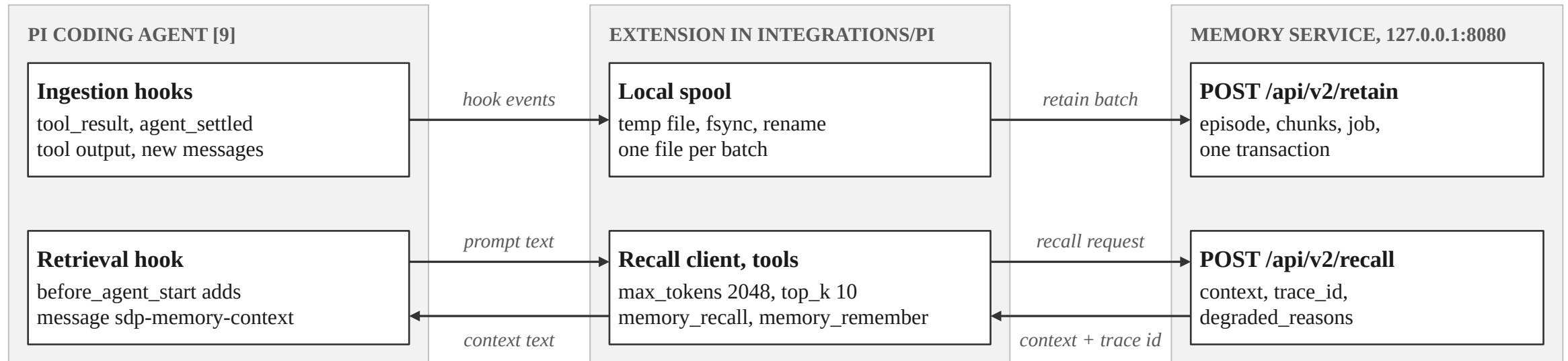
↓ *candidate ids, paths*

Graph arm in recall

Seeds: top 5 lexical, top 5 semantic, 10 by entity name. Two expansion rounds, 200 rows at most. PostgreSQL checks every id again.

Every fact traces back to its exact quote, chunk and raw event. Graph results are ranked by Neo4j row order, and we have not measured what the graph search adds.

PI INTEGRATION



What	How it works
Namespace	One per user, user:<login>, shared by all directories
Retention	New messages only, tracked by a cursor entry
Automatic recall	Date planner off, 15 s timeout, raw text included, max_distance 0.45
Tests	Completed, 9 TypeScript tests

Recall cutoff: Completed

Automatic recall passes max_distance 0.45. The memory_recall tool passes none.

INSTALL, PRIVACY AND CHECKS

- Not decided yet: a global install with pi install, and whether every session may be retained.
- Not retained: the injected message and the results of the memory tools.
- Command /memory-status flushes the spool and shows job state.
- Scripted check, 14 checks per run: 14 of 14 in four runs (claude-haiku-4-5 or claude-sonnet-5-5 through Pi). One fact, one machine.

ALGORITHMS

1 WRITE PATH: EXTRACT AND RESOLVE

```
extract(episode):
    prompt = rules + 80 newest current facts + events
    for attempt in 1..3: (call, then 2 repairs)
        claims = model(prompt)
        error = validate(claims) (schema, quotes, ids)
        if error: prompt += claims + error, else stop
    if error: fail the job (backoff, 5 attempts)
    embed missing chunks, then claim statements
    for claim in claims: (one transaction)
        slot = subject_id :: predicate
        start = claim.valid_from, else first cited event time
        same slot and value: add quote, keep status
        multiple, event, or no old row: insert as active
        single, correction, confidence >= 0.8:
            old: superseded, valid_to = max(old.valid_from, start)
            new: active, edge new supersedes old
        single otherwise: old and new become contested
        old row left active: mark dependants stale
    consolidate job: observation needs 2+ active supports
```

2 READ PATH: RECALL

```
recall(query, as_of, max_tokens = 8192, top_k = 20):
    if time words in query: ask model for a date window
    lexical = full text search (valid rows, 40)
    semantic = nearest(embed(query)) (valid rows, 40)
    temporal = newest in window (if a window is set)
    graph = expand(seeds), keep valid ids
    if max_distance: drop far vector hits
    if all arms empty, or extraction pending, or include_raw:
        add raw chunk matches
    score each id by RRF, sort by score, then by id
    used = 0, blocks = 0
    for block in ranked:
        if blocks == top_k: stop
        if used + tokens(block) <= max_tokens:
            add block, used += tokens(block), blocks += 1
    return context, trace_id
```

Fusion and packing

score(d) = sum over arms i of $1 / (60 + \text{rank}_i(d))$ [12]
 tokens(block) = $\text{ceil}(\text{characters} / 4)$

Why k = 60

The constant from the cited work. Fixed in code here, not tuned.

What validation proves

Each quote occurs word for word in the event it cites.
 It may still not support the claim.

TOOLS AND DATASET

Tool	Version	Actual role	Runs
Rust, Tokio, Axum, SQLx	1.96.1, 1.53.1, 0.8.9, 0.9.0	Service, async runtime, REST API and database access	Local
PostgreSQL, pgvector	17, pgvector 0.8.6	Authoritative store, full-text search [10] and HNSW cosine index [11]. Compose pins the floating tag only	Local, Docker
Neo4j	5.26	Graph projection over the HTTP endpoint, marked deprecated in 5.26 [8]	Local, Docker
Ollama	0.13.5	Qwen3-Embedding-0.6B, 1024-d [13]. Optional Qwen2.5-14B extractor: In progress, not usable yet	Local
Pi coding agent	1.0.2 locked, 0.85.0 installed	Hosts the memory extension and runs the default extraction worker [9]. Completed	Local agent, hosted model
Azure OpenAI	text-embedding-3-small	Extractor and embedder behind the same traits. In progress: written, not in the main code yet	Hosted
Qwen3-1.7B student	4-bit NF4 QLoRA [14]	Offline pipeline: In progress, not wired in. Serving through Ollama: Planned	Local, once served

Development and demo data

Namespace user:dev: the project's own documents and source, 79 batches through POST /api/v2/retain, plus invented example facts from four Pi recordings.

Scripted Pi check: one fact told, corrected and asked from fresh sessions, 14 checks per run.

Extraction-training data

784 windows and 3,706 memories from 10 LoCoMo conversations [15]. Train: 642 windows from 8. Held out: 142 from 2. Labelled once by a hosted teacher.

LoCoMo is CC BY-NC 4.0. We still have to confirm that keeping its turns in the repository is allowed.

Evaluation benchmarks

LongMemEval-S [2], September 2025 cleaned release: 500 questions, 23,867 episodes, 277 MB, pinned by SHA-256, not in Git. GPT-4o scored 60.6% on the S histories in the original paper release.

LoCoMo: 10 conversations, 1,986 questions. Planned secondary benchmark.

OUTPUT SCREENSHOTS

```

user
Remember this for later: the iOS signing certificate expires on March 3,
the design review happens every Tuesday at 11:00, and Kavya owns the
payments dashboard.

memory_remember
Evidence submitted for processing.

[CTX-OK] I've saved all three items to memory:
- The iOS signing certificate expires on March 3.
- The design review happens every Tuesday at 11:00.
- Kavya owns the payments dashboard.

Memory processes claims asynchronously, so I can't yet confirm they're
stored.

(Took 4s)

76 J196 R44k W44k CH99.7% $8.121 4.4%/1.0M (au claude-sonnet-5.5 • medium)
~/code/harness/second-review/sessions/payments-api 13088.6 tok/s

```

Figure 1. Session 1 in payments-api, 2026-10-06. The user tells Pi three example facts: the iOS signing certificate expires on March 3, the design review is every Tuesday at 11:00, and Kavya owns the payments dashboard. Pi calls memory_remember, which answers "Evidence submitted for processing", and replies that it saved all three items but cannot yet confirm they are stored. The CTX-OK marker comes from the presenter's own Pi instructions, not from the memory system.

```

mobile-app $ ls
README.md
mobile-app $ pi -e $EXT

Ponytail loaded: full

0.0%/1.0M (auto) claude-sonnet-5.5 • medium
When does the iOS signing certificate expire, when is the design review,
and who owns the payments dashboard?
~/code/harness/second-review/sessions/mobile-app

```

Figure 2. Session 2 in mobile-app, a fresh start with no history: ls shows only README.md. The question about the certificate date, the design review and the dashboard owner is typed in the editor. The empty lower part of the terminal is cropped away.

```

review happens every Tuesday at 11:00. Kavya owns the payments dashboard.
[136436e2-27fa-4b06-a62d-63646311cc04] The iOS signing certificate expires
on March 3. [fact: active; valid_from:Some(2026-10-06T01:05:12.189Z);
valid_to=None; event_at=None]
Evidence (01a10ebd-f7e5-7596-b871-3b4e41ce4dd2, assistant, 2026-10-06
01:05:12.189 UTC): iOS signing certificate expires on March 3.
[112cb98b-62f2-45c3-a556-37b17357f984] Kavya owns the payments dashboard.
[fact: active; valid_from:Some(2026-10-06T01:05:12.189Z); valid_to=None;
event_at=None]
Evidence (01a10ebd-f7e5-7596-b871-3b4e41ce4dd2, assistant, 2026-10-06
01:05:12.189 UTC): Kavya owns the payments dashboard.

[CTX-OK] This comes from memory of an earlier session. I haven't checked it
against any current files.

- iOS signing certificate: expires on March 3. The memory doesn't give a
year.
- Design review: every Tuesday at 11:00.
- Payments dashboard owner: Kavya.

(Took 2s)

74 J105 R30k W35k CH67.2% $8.044 4.5%/1.0M (au claude-sonnet-5.5 • medium)
~/code/harness/second-review/sessions/mobile-app 147.1 tok/s

```

Figure 3. Session 2 after the answer. The upper part is the end of the memory the extension injected from session 1, so its first line starts mid-sentence: the facts, each with status active and an evidence line. Below it the answer gives March 3 (no year), every Tuesday at 11:00 and Kavya.

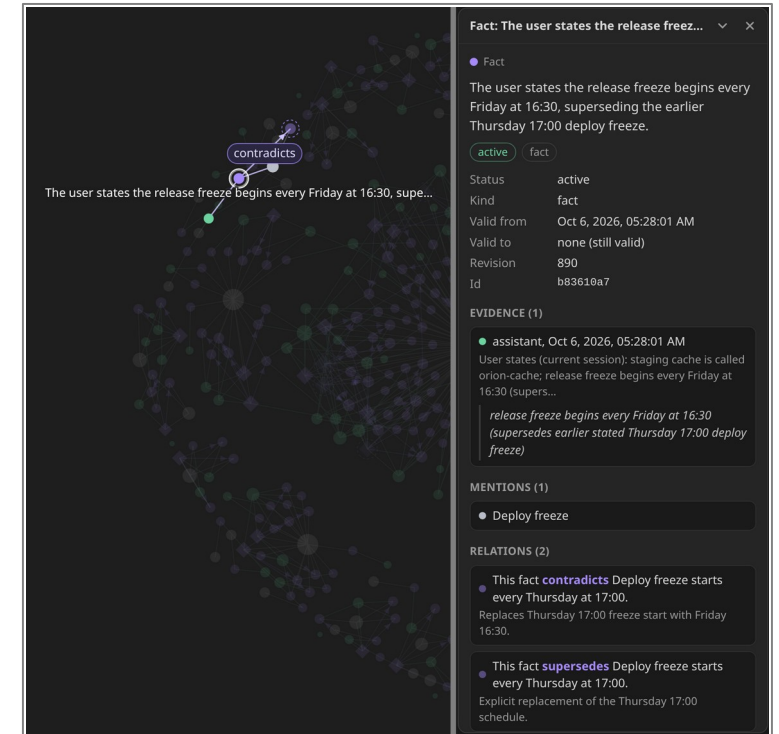


Figure 4. Graph page, user:dev, 2026-10-06. Two crops of one capture, joined at the grey line. Left: the selected fact (white ring), a contradicts relation and the superseded Thursday 17:00 fact (dashed ring). Right: its details panel, with status active, the evidence quote, the entity Deploy freeze and the relations contradicts and supersedes. Invented example facts.

RESULTS AND OBSERVATIONS

Measured finding	Result	Limit kept with it
Retrieval latency and context size	Date planning off: p50 58 ms, p95 72 ms, p99 75 ms at 10 requests per second (300 requests, 0 errors). With the planner: p95 1,870 ms, because 4 of 40 questions start a model call of 1.6 to 2.1 s. Median context 1,942 tokens, p95 2,041	One machine, local embeddings, 323 facts. Not tried on a large namespace. Tokens are characters divided by four
Provenance and version audit	323 of 323 facts have a source with a quote found word for word in its chunk. 0 of 106 single-value slots hold two active facts. Memory at idle: service 35 MB, PostgreSQL 224 MiB, Neo4j 1.27 GiB	Run on user:dev only, where just 2 facts are superseded, so it is not a held-out update set
Pi sessions, recording and scripted check	Session 2 (claude-sonnet-5-5), in another directory with no history, answered all three example facts. The scripted check passed 14 of 14 in four configurations	One scenario, invented facts, one machine, run by us. Shows the path works, not answer quality
Recall cutoff, 80 queries	Target fact in the context: 39 of 55 with no cutoff, 49 at 0.45. Follow-ups and small talk: mean context 6,925 to 1,250 characters	The same queries tuned the cutoff, so the claim is weak. Short prompts such as “ok” still pull in 3 to 5 kB
Local 14B extraction, 10 batches	Validation passed on 2 of 10 batches, the hosted model on 10 of 10. Before the context fix Ollama cut 15 of 18 prompts at 4,096 tokens, now 16,384 in and 4,096 out. GPU memory 10.6 GB at 4,096 and 12.9 GB at 16,384	One machine, one model, no repeats, stopped at 10 of 18 batches. Not usable for extraction yet
Live graph, user:dev	Neo4j projection: 382 nodes, 473 relationships. 133 of the 147 memories support at least one fact. Graph API counts equal direct Cypher and SQL counts, about 0.10 s per call at limit 200	One user, invented example facts. Chrome only, untested above a few thousand nodes
Automated tests	scripts/test.sh exits 0: Rust 83 (45 unit, 24 evidence_graph, 13 ollama_context, 1 postgres_api), Python 5, TypeScript 9	Database tests need TEST_DATABASE_URL and TEST_NEO4J_URL. Three live Ollama tests are opt-in and passed once
Offline agreement, distilled student	136 held-out windows. Relaxed F1 0.398 for the student and 0.000 for the base model (same type and subject, statement word overlap at least 0.5)	Agreement with the teacher, not accuracy. Student p50 6,115 ms against 2,868 ms in an unbatched run. Not wired in
LongMemEval-S	Not run	Needs a 277 MB dataset and a lot of model use. Until it runs, the effect of memory on answers is unmeasured

STATUS AND PENDING WORK

Module	Implemented functionality	Available evidence	Pending work	Status
M1 Service and persistence	REST API (8 V1, 13 V2 routes), console, PostgreSQL store, job queue	src/api.rs, knowledge_store.rs; 83 Rust tests	Authentication and hard delete: Status to confirm	Completed
M2 Extraction and embeddings	JSON claims, exact-quote checks, two repairs; Ollama embeddings, 1024-d	src/worker.rs, providers.rs. Pi check: 14 of 14	Extraction quality of the Pi worker is unmeasured	Completed
Local 14B extraction	Qwen2.5-14B in Ollama, 16,384 token context	Passed validation on 2 of 10 test batches, hosted 10 of 10	Run the other 8 batches. Planned: constrained decoding, fewer related ids, a larger repair budget	In progress
M3 Hybrid retrieval and graph	Four searches, RRF k = 60, validity filter, token packing, Neo4j projection, live graph page, distance cutoff	src/v2.rs, graph.rs, tests/evidence_graph.rs; latency on slide 14	Shorter context, LongMemEval-S recall, graph ablation, a rule for short prompts. A derives writer and a move off the deprecated Neo4j HTTP endpoint: Status to confirm	Completed
M4 Maintenance and evaluation	Retries, heartbeats, consolidation, traces, /healthz, /metrics, runner	src/worker.rs, benchmark/run.py, 5 protocol tests	Finish a first LongMemEval-S run, add p95 latency to the runner, run a human audit (Planned). Decay, re-embedding, backup: none exists	Completed. Benchmark run: In progress
Pi integration	Hooks, local spool, memory tools, injected context message	integrations/pi/, 9 tests, scripted check 14 of 14	Global install and the privacy of retaining all sessions: Status to confirm	Completed
Student extractor, offline	QLoRA of Qwen3-1.7B on 642 windows; 136 held-out windows scored	slm-distill/, scores recomputed 2026-10-06	Not wired in: merge the LoRA, convert to GGUF, ollama create, decide the schema, retest latency. Serving: Planned	In progress
Azure adapter and V1 console	AzureExtractor, AzureEmbedder, V1 console, browse and stats endpoints	Written: 12 files, 1,587 lines added	Rebase onto the V2 code (conflicts in src/main.rs and src/ui.html), then add a JsonModel adapter for the V2 worker	In progress

REFERENCES

- [1] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, "Lost in the middle: How language models use long contexts," *Trans. Assoc. Comput. Linguist.*, vol. 12, pp. 157-173, 2024. [Online]. Available: <https://aclanthology.org/2024.tacl-1.9/>
- [2] D. Wu, H. Wang, W. Yu, Y. Zhang, K.-W. Chang, and D. Yu, "LongMemEval: Benchmarking chat assistants on long-term interactive memory," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025, pp. 86809-86836. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2025/hash/d813d324dbf0598bbdc9c8e79740ed01-Abstract-Conference.html. Also arXiv:2410.10813.
- [3] C. Packer, S. Wooders, K. Lin, V. Fang, S. G. Patil, I. Stoica, and J. E. Gonzalez, "MemGPT: Towards LLMs as operating systems," arXiv preprint arXiv:2310.08560, Oct. 2023 (rev. Feb. 2024). [Online]. Available: <https://arxiv.org/abs/2310.08560>
- [4] P. Rasmussen, P. Paliychuk, T. Beauvais, J. Ryan, and D. Chalef, "Zep: A temporal knowledge graph architecture for agent memory," arXiv preprint arXiv:2501.13956 [cs.CL], Jan. 2025. [Online]. Available: <https://arxiv.org/abs/2501.13956>
- [5] P. Chhikara, D. Khant, S. Aryan, T. Singh, and D. Yadav, "Mem0: Building production-ready AI agents with scalable long-term memory," in *Proc. 28th Eur. Conf. Artif. Intell. (ECAI)*, Oct. 2025, ser. *Frontiers in Artificial Intelligence and Applications*, vol. 413. IOS Press, 2025, pp. 2993-3000, doi: 10.3233/FAIA251160. Also arXiv:2504.19413 [cs.CL].
- [6] S. Daga, S. Sreedhar, and D. Shah, "LongMemEval," supermemory research (vendor-hosted evaluation report on LongMemEval-S, not peer reviewed), May 2026. [Online]. Available: <https://supermemory.ai/research/longmembench/> (accessed Oct. 6, 2026).
- [7] C. Latimer, N. Boschi, A. Neeser, C. Bartholomew, G. Srivastava, X. Wang, and N. Ramakrishnan, "Hindsight is 20/20: Building agent memory that retains, recalls, and reflects," arXiv preprint arXiv:2512.12818 [cs.CL], Dec. 2025. [Online]. Available: <https://arxiv.org/abs/2512.12818>
- [8] Neo4j, Inc., "Query the database," in *Neo4j HTTP API Manual*, Version 5 (page marked deprecated in 5.26). Neo4j, Inc., 2026. [Online]. Available: <https://neo4j.com/docs/http-api/5/query/> (accessed Oct. 6, 2026).
- [9] M. Zechner et al., "pi: AI agent toolkit (unified LLM API, agent loop, TUI, coding agent CLI)," package @earendil-works/pi-coding-agent, v1.0.2 (version locked by the project's package-lock.json; latest is 1.0.4 on Oct. 6, 2026), GitHub repository, 2026. [Online]. Available: <https://github.com/earendil-works/pi> (accessed Oct. 6, 2026).
- [10] PostgreSQL Global Development Group, "PostgreSQL 17 Documentation, Chapter 12: Full Text Search (Sections 12.2.2, 12.3.2, 12.3.3, 12.9) and Section 64.4: GIN Indexes," PostgreSQL 17.11 documentation, 2026. [Online]. Available: <https://www.postgresql.org/docs/17/textsearch.html> (accessed Oct. 6, 2026).
- [11] A. Kane et al., "pgvector: Open-source vector similarity search for Postgres," source code repository and README (0.8.6 installed in the test database, v0.8.7 tag also released), Oct. 2026. [Online]. Available: <https://github.com/pgvector/pgvector> (accessed Oct. 6, 2026).
- [12] G. V. Cormack, C. L. A. Clarke, and S. Büttcher, "Reciprocal rank fusion outperforms Condorcet and individual rank learning methods," in *Proc. 32nd Int. ACM SIGIR Conf. Research and Development in Information Retrieval (SIGIR '09)*, Boston, MA, USA, Jul. 2009, pp. 758-759, doi: 10.1145/1571941.1572114.
- [13] Y. Zhang, M. Li, D. Long, X. Zhang, H. Lin, B. Yang, P. Xie, A. Yang, D. Liu, J. Lin, F. Huang, and J. Zhou, "Qwen3 Embedding: Advancing text embedding and reranking through foundation models," arXiv preprint arXiv:2506.05176 [cs.CL], Jun. 2025. [Online]. Available: <https://arxiv.org/abs/2506.05176>
- [14] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," in *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023, pp. 10088-10115, doi: 10.52202/075280-0441. Also arXiv:2305.14314.
- [15] A. Maharana, D.-H. Lee, S. Tulyakov, M. Bansal, F. Barbieri, and Y. Fang, "Evaluating very long-term conversational memory of LLM agents," in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics (Volume 1: Long Papers)*, Bangkok, Thailand, Aug. 2024, pp. 13851-13870, doi: 10.18653/v1/2024.acl-long.747. [Online]. Available: <https://aclanthology.org/2024.acl-long.747/> (preprint: arXiv:2402.17753)